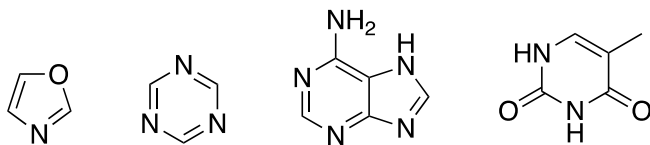
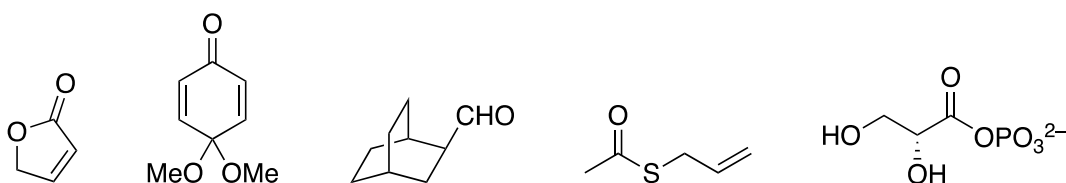


Redox Reactions of Alcohols and Carbonyl Compounds; Organometallics

1. Many heterocycles contain carbon atoms that are equivalent in oxidation level to carboxylic acid derivatives. In each of the heterocycles below, circle the atom(s) at the same oxidation level as a carboxylic acid.

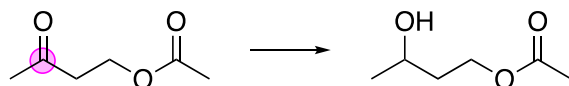


2. Indicate the oxidation numbers of the carbonyl carbons in each of the compounds below.

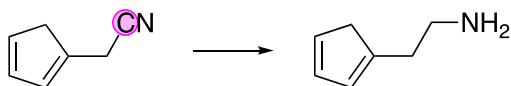


3. Classify each reaction below as oxidation or reduction at the indicated carbon, and then add reagents above the arrow that could be used to effect the transformation.

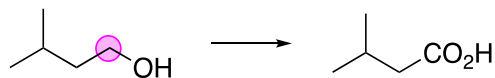
A.



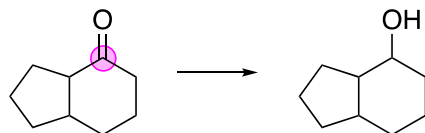
B.



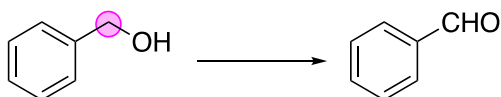
C.



D.

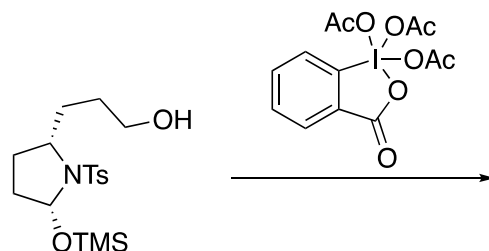


E.

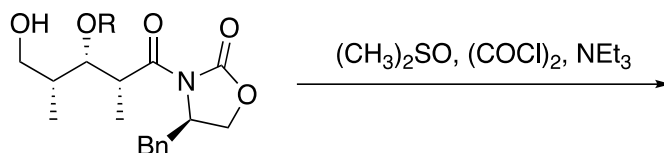


4. Provide the neutral organic product of each reaction after appropriate workup. If you expect a reaction to yield multiple isomers, draw only the major product.

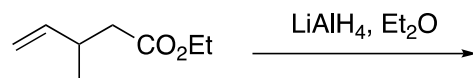
A.



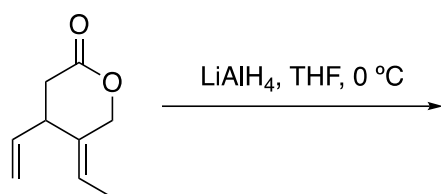
B.



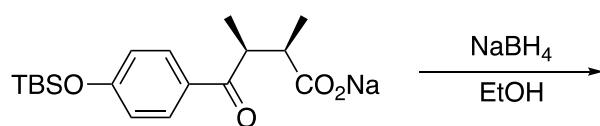
C.



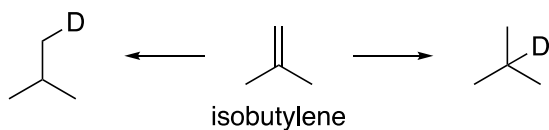
D.



E.



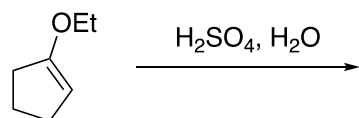
5. For a study of the bond dissociation energies of carbon–deuterium bonds, the following two deuterated hydrocarbons were needed. Devise syntheses of these two compounds starting from isobutylene and using inorganic reagents of your choice (including D_2O).



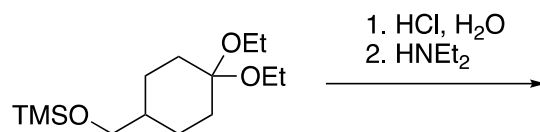
Nucleophilic Additions to Carbonyl Compounds

6. Provide the neutral organic product of each reaction after appropriate workup. If you expect a reaction to yield multiple isomers, draw only the major product.

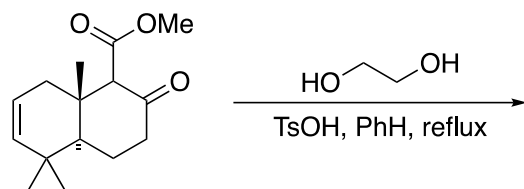
A.



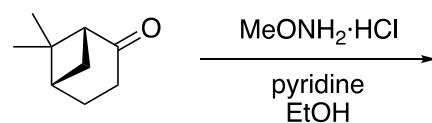
B.



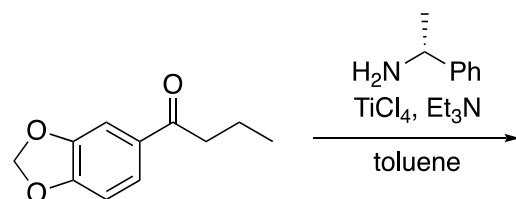
C.



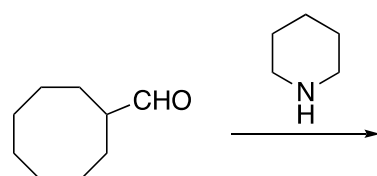
D.



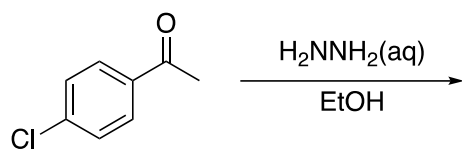
E.



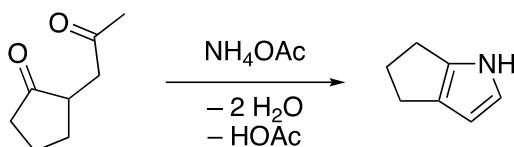
F.



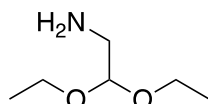
G.



7. Ammonium acetate reacts with 1,4-diones to yield *N*-unsubstituted pyrroles. Draw an abbreviated mechanism for this reaction. Begin by drawing an initially formed enamine intermediate, then use curved arrows to show how this intermediate cyclizes under acid catalysis with loss of water. Label each step as one of the ten elementary steps of polar organic reaction mechanisms.



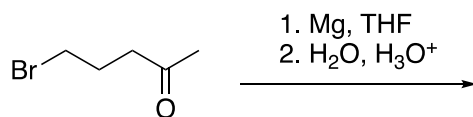
8. 2-Aminoacetaldehyde diethyl acetal is a useful building block for the construction of certain aromatic heterocycles. Devise a synthesis of this compound from starting materials that contribute no more than one carbon each to the target, keeping the following facts in mind:
(1) 2-Aminoacetaldehyde decomposes immediately upon preparation and cannot be isolated.
(2) Formyl chloride is not a stable compound but its ester and anhydride analogues are stable enough to carry on.



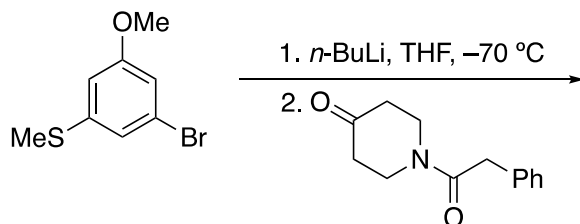
2-aminoacetaldehyde diethyl acetal

9. Provide the neutral organic product of each reaction after appropriate workup. If you expect a reaction to yield multiple isomers, draw only the major product.

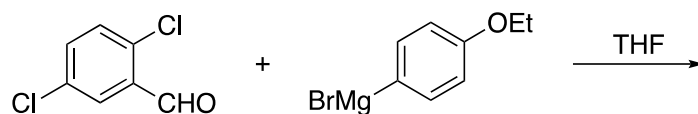
A.



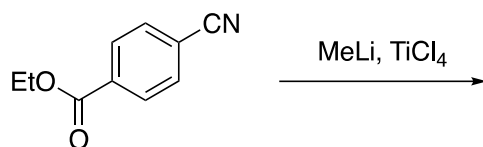
B.



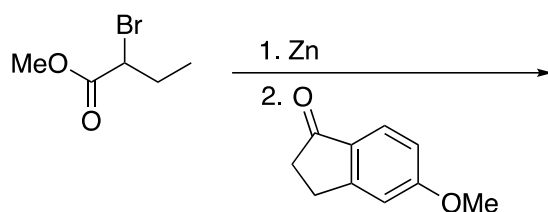
C.



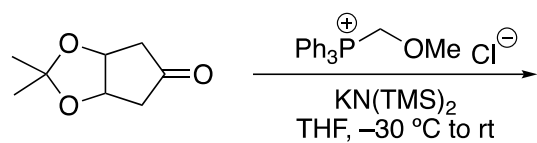
D.



E.

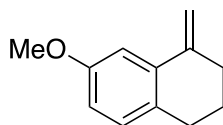


F.

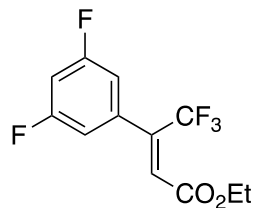


10. For each of the following alkenes, draw the structure of a phosphonium ylide and ketohyde that could be combined to synthesize the alkene via a Wittig reaction.

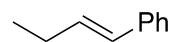
A.



B.

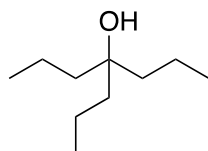


C.

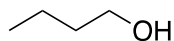


11. Starting from alcohols containing no more than three carbon atoms and inorganic reagents of your choice, devise syntheses of the following compounds.

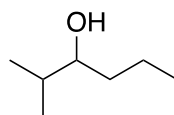
A.



B.



C.



D.

